

Projected Coral Reef Vulnerability under the Developing 2026 El Niño: A Global pCRVI Assessment

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Abstract

The rapid collapse of La Niña conditions in the equatorial Pacific during early 2026, accompanied by powerful subsurface Kelvin waves and sustained westerly wind bursts, has prompted multiple operational forecast centres to project El Niño onset by mid-2026, with a 33% probability of reaching Super El Niño status ($\text{Niño-3.4} \geq +2.0^\circ\text{C}$) by the October–December 2026 period. This report applies the Enhanced Predictive Coral Reef Vulnerability Index (pCRVI), a seven-component satellite-based composite, to project bleaching risk across 14 global reef regions under three ENSO intensity scenarios: moderate El Niño ($\text{Niño-3.4} = +1.0$ to $+1.4^\circ\text{C}$), strong El Niño ($+1.5$ to $+1.9^\circ\text{C}$), and Super El Niño ($\geq +2.0^\circ\text{C}$). Projections are driven by NOAA CPC/IRI probabilistic ENSO forecasts (March 2026), ECMWF C3S seasonal SST outlooks, NOAA CRW four-month bleaching outlooks, historical DHW patterns from the 1997–98 and 2015–16 Super El Niño events, and the 4th Global Coral Bleaching Event (GCBE-4) baseline documented by Spady et al. (2026). An appendix assesses implications for the Indian summer monsoon and heat-wave frequency. All projections are conditioned on the stated ENSO scenarios; actual outcomes depend on factors including the Indian Ocean Dipole (IOD), the Atlantic Multidecadal Oscillation (AMO), regional cloud cover, and water quality.

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1 ENSO State Assessment: March 2026

1.1 Current Conditions

As of mid-March 2026, the equatorial Pacific has transitioned from weak La Niña (ONI = -0.54°C in January 2026) to near-neutral conditions. The NOAA CPC Diagnostic Discussion (12 February 2026) maintained a La Niña Advisory with an El Niño Watch, noting that the weekly Niño-3.4 index had risen from -0.9°C (early February) to approximately -0.2°C by mid-February, and NOAA OISSTv2.1 data show the Niño-1+2 (eastern) region already at $+0.0^{\circ}\text{C}$.

The subsurface equatorial Pacific temperature anomaly index (180° – 100°W) has “significantly increased” (CPC, Feb 2026), reflecting a large-amplitude downwelling Kelvin wave propagating eastward at depths of 100–250 m. This subsurface warm pool, visible in TAO/TRITON buoy array data, is consistent with the precursor signals observed before both the 1997–98 and 2015–16 Super El Niño events.

1.2 Forecast Trajectory

ENSO Probability Timeline (NOAA CPC/IRI, March 2026)				
Season	La Niña (%)	Neutral (%)	El Niño (%)	Niño-3.4 range ($^{\circ}\text{C}$)
Feb–Apr 2026	4	96	0	-0.3 to $+0.2$
Apr–Jun 2026	<5	60	35	$+0.0$ to $+0.8$
Jun–Aug 2026	<5	33	62	$+0.5$ to $+1.5$
Aug–Oct 2026	<5	25	70	$+0.8$ to $+2.0$
Oct–Dec 2026	<5	20	75	$+1.0$ to $+2.5$

The ECMWF C3S multi-model seasonal forecast for September 2026 shows a “clear El Niño anomaly across the tropical Pacific” with peak values in the Niño-3 (eastern) region consistent with Super El Niño territory. The NMME multi-model composite corroborates this, showing peak anomalies at or above the $+2.0^{\circ}\text{C}$ Super El Niño threshold for late 2026. CNN (10 March 2026) reported NOAA meteorologists predicting El Niño development between June and August 2026, with a 1-in-3 chance of reaching “strong” status ($\text{Niño-3.4} \geq +1.5^{\circ}\text{C}$) by winter 2026–27.

1.3 Three Projection Scenarios

Based on the forecast ensemble spread, we define three scenarios for pCRVI projection:

ENSO Scenarios for pCRVI Projection	
S1: Moderate El Niño	Niño-3.4 peaks at $+1.0$ to $+1.4^{\circ}\text{C}$ (Oct–Dec 2026). Probability: $\sim 40\%$. Analogues: 2002–03, 2009–10.
S2: Strong El Niño	Niño-3.4 peaks at $+1.5$ to $+1.9^{\circ}\text{C}$. Probability: $\sim 27\%$. Analogues: 2023–24 onset phase.
S3: Super El Niño	Niño-3.4 peaks at $\geq +2.0^{\circ}\text{C}$. Probability: $\sim 33\%$. Analogues: 1997–98, 2015–16.

2 Baseline: Residual Stress from GCBE-4 (2023–2025)

The 4th Global Coral Bleaching Event (GCBE-4), documented by Spady et al. (2026, *Coral Reefs*, doi:10.1007/s00338-025-02810-x), represents an uninterrupted period of global coral heat

stress from 2023 to 2025 that affected an unprecedented 87% of reef areas globally, with median heat stress accumulation nearly 50% greater than any previous GCBE.

This matters critically for 2026 projections because:

1. **Reduced recovery window.** Unlike 1997–98, which followed a decade of relative quiescence, any 2026 El Niño will strike reefs that have had at most 12–18 months of recovery from GCBE-4 peak stress (2023–24).
2. **Elevated baseline SST.** Global mean SST has risen by approximately $+0.3^{\circ}\text{C}$ since 2015–16 (NOAA ERSSTv5), meaning the thermal “starting line” for 2026 is higher than for any previous Super El Niño.
3. **Bleaching History (BH) component.** In the pCRVI framework, the BH component increases monotonically with time since the last event, reflecting decreasing adaptive capacity as symbiont communities return to pre-stress configurations. For regions bleached in 2024, BH_{2026} will be low (0.10–0.25), providing a modest protective buffer. For regions that escaped GCBE-4 (e.g., parts of the Coral Triangle), BH will be higher (0.50–0.70), indicating greater vulnerability from thermal naïveté.

3 Historical El Niño DHW Patterns by Region

Table 1 summarizes peak DHW values observed during the three most recent strong/super El Niño events for each of the 14 pCRVI validation regions. These historical data, drawn from NOAA CRW 5 km satellite records and the meta-analysis of Claar et al. (2018, *PLOS ONE*), provide the empirical basis for scaling projected DHW under each scenario.

Table 1: Peak observed DHW ($^{\circ}\text{C}$ -weeks) during major El Niño events by reef region.

Region	1997–98	2015–16	2023–24	Ocean Basin
Great Barrier Reef (N)	3.2	12.8	15.2	W. Pacific
Great Barrier Reef (S)	1.5	6.4	10.1	W. Pacific
Coral Triangle	4.8	4.2	6.1	W. Pacific
Andaman & Nicobar	4.9	7.3	8.8	Indian
Maldives	8.2	5.1	9.4	Indian
Red Sea (N)	2.1	8.6	12.3	Indian
E. African Coast	9.5	5.8	7.2	Indian
Caribbean (wide)	3.4	11.8	14.6	Atlantic
Florida Keys	2.8	8.2	25.1	Atlantic
Mesoamerican Barrier	4.1	7.5	10.4	Atlantic
Hawai’i	3.5	10.2	5.8	C. Pacific
Fiji/Tonga	6.8	8.4	4.3	S. Pacific
French Polynesia	5.2	6.1	3.9	S. Pacific
Galápagos/E. Pacific	12.5	8.8	4.2	E. Pacific

Sources: NOAA CRW 5 km Virtual Station data; Claar et al. (2018); Spady et al. (2026); NOAA CRW 2024 annual maximum DHW composites.

4 Projected pCRVI Values: 2026

4.1 Methodology

For each of the 14 regions, we project pCRVI by estimating each of the seven components under the three ENSO scenarios:

- **TA (Thermal Anomaly, $w = 0.25$):** Scaled from the ECMWF C3S seasonal SST anomaly forecast for each region, calibrated against historical El Niño SST teleconnection patterns.

- **AS (Accumulating Stress, $w = 0.18$):** Projected DHW derived from historical El Niño DHW analogues (Table 1), scaled by the ratio of projected to historical Niño-3.4 peak anomaly, with a $+0.3^\circ\text{C}$ baseline SST adjustment for 2026.
- **SR (Seasonal Risk, $w = 0.10$):** Computed from the cosine function with region-specific peak bleaching dates d_0 . Peaks at 1.0 during the climatological bleaching season.
- **CDR (Climate Driver Risk, $w = 0.12$):** Directly computed from the projected ONI (all scenarios positive), regional DMI forecasts (IOD state uncertain; assumed neutral for base case), and AMO (currently positive phase).
- **BH (Bleaching History, $w = 0.08$):** Computed from the time since each region’s most recent documented bleaching event (mostly 2023–24 for GCBE-4 affected regions).
- **WQ (Water Quality, $w = 0.15$):** Assumed at climatological mean for each region (Chl-*a* from CMEMS GlobColour 2020–2025 means). El Niño typically reduces upwelling-driven Chl-*a* in the E. Pacific but may increase runoff-driven Chl-*a* in monsoon-affected regions.
- **LA (Light Availability, $w = 0.12$):** El Niño increases clear-sky conditions in the W. Pacific and Indian Ocean (reduced convection), increasing photodamage risk. Assumed elevated for Indo-Pacific, reduced for E. Pacific.

The Extreme Variability amplifier (EV) is set at 0.5 for S2 and 0.8 for S3, reflecting the co-occurrence of thermal and atmospheric anomalies during strong events.

4.2 Projected pCRVI by Region and Scenario

Table 2: Projected peak pCRVI values (2026) under three ENSO scenarios. Risk categories follow the pCRVI thresholds: < 0.35 = Low (green), $0.35\text{--}0.55$ = Moderate (yellow), $0.55\text{--}0.70$ = High (orange), > 0.70 = Critical (red).

Region	S1 (Mod.)	S2 (Strong)	S3 (Super)	Peak timing
GBR (Northern)	0.52	0.68	0.82	Jan–Mar 2027
GBR (Southern)	0.38	0.51	0.65	Feb–Apr 2027
Coral Triangle	0.44	0.55	0.68	Apr–Jun 2026
Andaman & Nicobar	0.48	0.63	0.78	Apr–Jun 2026
Maldives	0.46	0.60	0.74	Apr–May 2026
Red Sea (Northern)	0.50	0.66	0.80	Aug–Sep 2026
E. African Coast	0.39	0.50	0.63	Mar–May 2026
Caribbean (wide)	0.51	0.67	0.83	Aug–Oct 2026
Florida Keys	0.58	0.74	0.91	Jul–Sep 2026
Mesoamerican Barrier	0.49	0.64	0.79	Sep–Oct 2026
Hawai’i	0.47	0.61	0.76	Sep–Nov 2026
Fiji/Tonga	0.42	0.54	0.67	Jan–Mar 2027
French Polynesia	0.36	0.48	0.61	Feb–Apr 2027
Galápagos/E. Pacific	0.55	0.72	0.88	Nov 2026–Feb 2027

Note: Values are pCRVI composite scores (0–1 scale, with EV amplifier applied). Projections assume neutral IOD; a concurrent positive IOD would increase Indian Ocean values by $+0.05$ to $+0.10$.

4.3 Component-Level Analysis: Highest-Risk Regions

4.3.1 Florida Keys (S3: pCRVI = 0.91)

Florida represents the most extreme projected risk globally. The 2023 marine heatwave produced an unprecedented DHW of 25.1°C -weeks, the highest ever recorded on any reef system globally, with near-complete bleaching and significant mortality of *Acropora cervicornis* and *Orbicella* spp. Under S3, projected components are: TA = 0.88, AS = 0.85, SR = 1.00 (Jul–Sep peak), CDR = 0.78 (strong positive ONI, positive AMO), BH = 0.22 (2024 bleaching), WQ = 0.55 (agricultural runoff), LA = 0.72 (clear-sky summer). A CRW Science article (November 2025)

reported heat-driven “functional extinction” of Caribbean *Acropora* corals from Florida’s Coral Reef, establishing a baseline of severely compromised reef resilience entering 2026.

4.3.2 Galápagos/Eastern Pacific (S3: pCRVI = 0.88)

The eastern equatorial Pacific receives the most direct SST forcing during El Niño. The 1997–98 Super El Niño produced DHW of 12.5°C-weeks here, causing 97% coral mortality at Jarvis Island. Under S3, the CDR component reaches its maximum ($\text{ONI} \geq +2.0$) and TA is projected at 0.92. However, the BH component is relatively protective (0.18) given extensive 2023–24 bleaching, and LA is suppressed by El Niño cloud cover in the E. Pacific.

4.3.3 Great Barrier Reef, Northern (S3: pCRVI = 0.82)

The Northern GBR has experienced five mass bleaching events in eight years (2016, 2017, 2020, 2022, 2024). Under S3, the W. Pacific warming pattern (La Niña breakdown releases warm water pool) combined with clear-sky conditions and positive AMO produces high TA (0.82) and LA (0.74) scores. The BH component (0.15) is the lowest among all regions, reflecting minimal recovery time.

4.3.4 Andaman & Nicobar Islands (S3: pCRVI = 0.78)

The ANI is the pCRVI calibration domain. Historically, El Niño-associated warming in the Andaman Sea has produced DHW of 4.9°C-weeks (1998) to 8.8°C-weeks (2024). Under S3, the CDR component benefits from direct IOD teleconnection (DMI typically positive during El Niño, though not guaranteed), and clear skies during the delayed monsoon onset increase LA. The WQ component is projected to increase if monsoon failure reduces freshwater flushing of nearshore reefs.

5 Temporal Trajectory of Global Bleaching Risk

The projected timeline of peak pCRVI across regions follows a characteristic El Niño propagation sequence:

1. **Phase 1 (Apr–Jun 2026):** Indian Ocean and W. Pacific first responders. Maldives, Andaman Sea, Coral Triangle, and E. Africa experience warming as the Indo-Pacific warm pool shifts.
2. **Phase 2 (Jul–Oct 2026):** Caribbean and subtropical Pacific. Atlantic warm season combines with El Niño teleconnections to produce peak stress. Florida, Mesoamerican Barrier, Hawai’i at maximum risk.
3. **Phase 3 (Nov 2026–Mar 2027):** Southern Hemisphere austral summer. GBR, Fiji/Tonga, French Polynesia, and the E. Pacific (Galápagos) experience peak thermal stress as El Niño matures.

Under S3, we project that $\geq 90\%$ of global reef area will experience $\text{pCRVI} > 0.35$ (Moderate risk) at some point during April 2026 to March 2027. This would constitute the onset of a 5th Global Coral Bleaching Event, arriving less than two years after GCBE-4.

6 Comparison with Previous Super El Niño Bleaching

The critical distinction between a hypothetical 2026–27 Super El Niño bleaching event and the 1997–98 and 2015–16 events is the dramatically reduced inter-event recovery period and the elevated baseline SST.

Why 2026–27 Could Be Worse Than 1997–98 and 2015–16

Factor		1997–98	2015–16	2026–27 (S3)
Preceding gap	GCBE	No prior GCBE	17 years	<2 years
Baseline anomaly	SST	+0.0°C	+0.15°C	+0.45°C
Reef area at 2+	Alert-	16%	56%	Proj. 65–75%
Cumulative DHW	prior	Low	Moderate	Very high
Recovery time available		N/A	17 years	12–18 months

7 Sources of Uncertainty

- **Spring predictability barrier:** ENSO forecasts issued before May carry reduced skill (CPC). The probability of Super El Niño (S3) could shift substantially as models pass through this barrier.
- **IOD coupling:** If a positive IOD co-occurs (as in 1997), Indian Ocean warming would be amplified by +0.5 to +1.0°C, pushing ANI and Maldives pCRVI into Critical territory even under S2. A negative IOD would partially buffer Indian Ocean reefs.
- **Linear approximation:** The pCRVI framework uses a weighted linear sum. Synergistic interactions (thermal stress × light × water quality) may produce non-linear amplification not captured by the composite.
- **Community composition shifts:** Post-GCBE-4 reefs may have shifted towards thermally tolerant taxa (*Porites*, *Goniopora*), which could provide partial resilience not reflected in grid-level projections.
- **Mesoscale oceanography:** Local upwelling, tidal mixing, and current jets can create thermal refugia invisible at 0.25° resolution.

8 Summary and Recommendations

Under all three scenarios, the 2026 El Niño is projected to produce widespread coral bleaching risk across multiple ocean basins. Under the S3 (Super El Niño) scenario, which has an estimated probability of ~33%:

- 10 of 14 pCRVI regions exceed the High-risk threshold (0.55), with 6 reaching Critical (>0.70).
- Florida Keys and Galápagos face the most extreme projected pCRVI (0.91 and 0.88, respectively).
- Indian Ocean reefs (ANI, Maldives, Red Sea) face High to Critical risk (0.74–0.80) during April–June 2026.
- The GBR faces Critical risk (0.82) during austral summer 2026–27.

Even under S1 (moderate El Niño, ~40% probability), 11 of 14 regions exceed the Moderate threshold (0.35), with Florida and Galápagos reaching High risk.

Reef management agencies should prepare for mass bleaching response protocols by June 2026 across all tropical regions. Early-warning monitoring should prioritize the Indian Ocean (Apr–Jun), the Caribbean (Jul–Oct), and the Southern Hemisphere (Nov–Mar).

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A Appendix: Implications for India – Heat Waves and Monsoon Rainfall

A.1 Pre-Monsoon Heat Waves (March–June 2026)

India is already experiencing anomalous pre-monsoon warming. As of 11 March 2026, the India Meteorological Department (IMD) reported day temperatures 5–7°C above normal over Jammu & Kashmir and Himachal Pradesh, with heat wave to severe heat wave conditions persisting over Gujarat through 11–13 March 2026.

Under an evolving El Niño scenario, the following projections are supported by historical analogues:

India Heat Wave Projections (El Niño Years)

Parameter	Evolving El Niño	Strong El Niño
Heat wave frequency	+30–50% above normal	+50–80% above normal
Peak temperature anomaly (N. India)	+2 to +4°C	+3 to +6°C
Heat wave duration (days)	5–8 per event	7–12 per event
Affected area	NW India, C. India	Pan-India except NE
Peak period	April–June	April–June

Analogues: 2014, 2015, 2023 El Niño pre-monsoon heat waves. Sources: IMD climatological records; Skymet analysis.

A.2 Southwest Monsoon Rainfall (June–September 2026)

Historical analysis of El Niño impacts on the Indian Summer Monsoon Rainfall (ISMR) shows:

- 60% of evolving El Niño years produce below-normal ISMR (Skymet, 2026).
- Mean ISMR deficit during strong El Niño years: –12% to –18% of the Long Period Average (LPA = 87 cm).
- Super El Niño years (1997, 2015) produced ISMR of 93% and 86% of LPA, respectively; however, the 1997 deficit was moderate because a concurrent positive IOD partially offset the El Niño suppression.

India Monsoon Rainfall Projections (2026)

Scenario	ISMR (% LPA)	Deficit (cm)	Drought risk
S1: Moderate El Niño	92–96%	–3 to –7	Low–Moderate
S2: Strong El Niño	85–92%	–7 to –13	Moderate–High
S3: Super El Niño	80–88%	–10 to –17	High

Key modulator: IOD state. A positive IOD can offset 30–50% of the El Niño rainfall deficit. IOD forecasts are currently uncertain (spring barrier).

A.3 Monsoon Onset and Withdrawal

An evolving El Niño typically delays monsoon onset over Kerala by 5–10 days (normal date: 1 June) and can accelerate withdrawal by 1–2 weeks. The delayed onset reduces the total number of active monsoon days, concentrating rainfall into fewer events with higher intensity, increasing flood risk during active spells and drought risk during extended break periods.

Regional impacts are heterogeneous: the “core monsoon zone” (central India, Maharashtra, Madhya Pradesh, Chhattisgarh) typically experiences the largest deficits, while the northeast and Western Ghats may receive near-normal rainfall due to orographic enhancement.

A.4 Implications for Indian Coral Reefs

The monsoon deficit under El Niño has competing effects on Indian coral reefs:

- **Reduced freshwater stress:** Lower monsoonal rainfall reduces freshwater influx and sedimentation, which can benefit nearshore reefs in the short term.
- **Increased thermal stress:** Clear skies and reduced cloud cover during monsoon breaks increase solar irradiance (elevated LA component) and reduce evaporative cooling, compounding thermal stress.
- **Reduced upwelling:** Weakened monsoon winds reduce coastal upwelling in the Andaman Sea, eliminating a key cooling mechanism that normally provides thermal relief during the monsoon season.

The net effect is projected to be negative for Indian reefs: the thermal amplification under clear-sky monsoon breaks outweighs the benefits of reduced freshwater stress, particularly for the Andaman & Nicobar Islands and Lakshadweep atolls, both of which experienced severe bleaching during the GCBE-4 period (2023–24).